

Optimization Techniques in Power System (BT EE 610)



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Tabu Search Algorithm

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History Of Tabu Search

- A very simple memory mechanism is described in Glover (1977) to implement the *oscillating assignment* heuristic
- Hansen . 1986. put The steepest ascent heuristic for coalition programming , Congress on numerical methods in coalition optimization .
- Glover (1986) introduces *tabu search* as a “meta-heuristic” To impose on another heuristic and Glover (1989) provide a full description of the method .

Introduction

- The word tabu (or taboo) comes from Tongan, a language of Polynesia, where it indicates things be touched because they are sacred.
- Now it also means “a prohibition imposed by social custom”.
- In TS, when finding the next solution to visit , some solution elements (or moves) are regarded as tabu , they cannot be used in constructing the next solution

Overview of Tabu search

- Tabu search is based on introducing **flexible memory** structures in conjunction with **strategic restrictions** and **aspiration levels** as a means for exploiting search spaces.
- Meta-heuristic that guides a local heuristic search procedure to explore the solution space beyond local optimum by use of a **Tabu list**.
- Originated from surrogate constraint methods and cutting plane approaches .

Overview Of Tabu Search

- Cutting plane: A cutting plane (or cut) for any IP is a new functional constraint that reduces the feasible region for the relaxation without eliminating any feasible solution for the IP problem.
- Used to solve fusion (finite solution set) optimization problems
- A dynamic neighborhood search method use of a flexible memory to restrict the next solution choice to some subset of neighborhood of current solution

Tabu Search Strategy

- **There are three main strategies :**
 - **Forbidding strategy:** control what enters to the tabu list.
 - **Freeing strategy:** control what exits the tabu list and when.
 - **Short-term strategy:** manage interplay between the forbidding strategy and freeing strategy to select trial solutions.

Parameters Of Tabu Search

- Local search procedure
- Neighborhood structure
- Aspiration conditions
- Form of tabu moves
- Addition of a tabu move
- Maximum size of tabu list
- Stopping rule

Basic components of Tabu Search

- A chief way to exploit memory in tabu search is to classify a subset of the moves in a neighborhood as forbidden (or **tabu**).
- A **neighborhood** is constructed to identify adjacent solutions that can be reached from current solution.
- The classification depends on the history of the search, and specially on the recency or frequency that certain move or solution components, called **attributes**, have participated in generating past solutions .
- A **tabu list** records forbidden moves, which are referred to as **tabu moves** .

Basic Tabu Search Algorithm

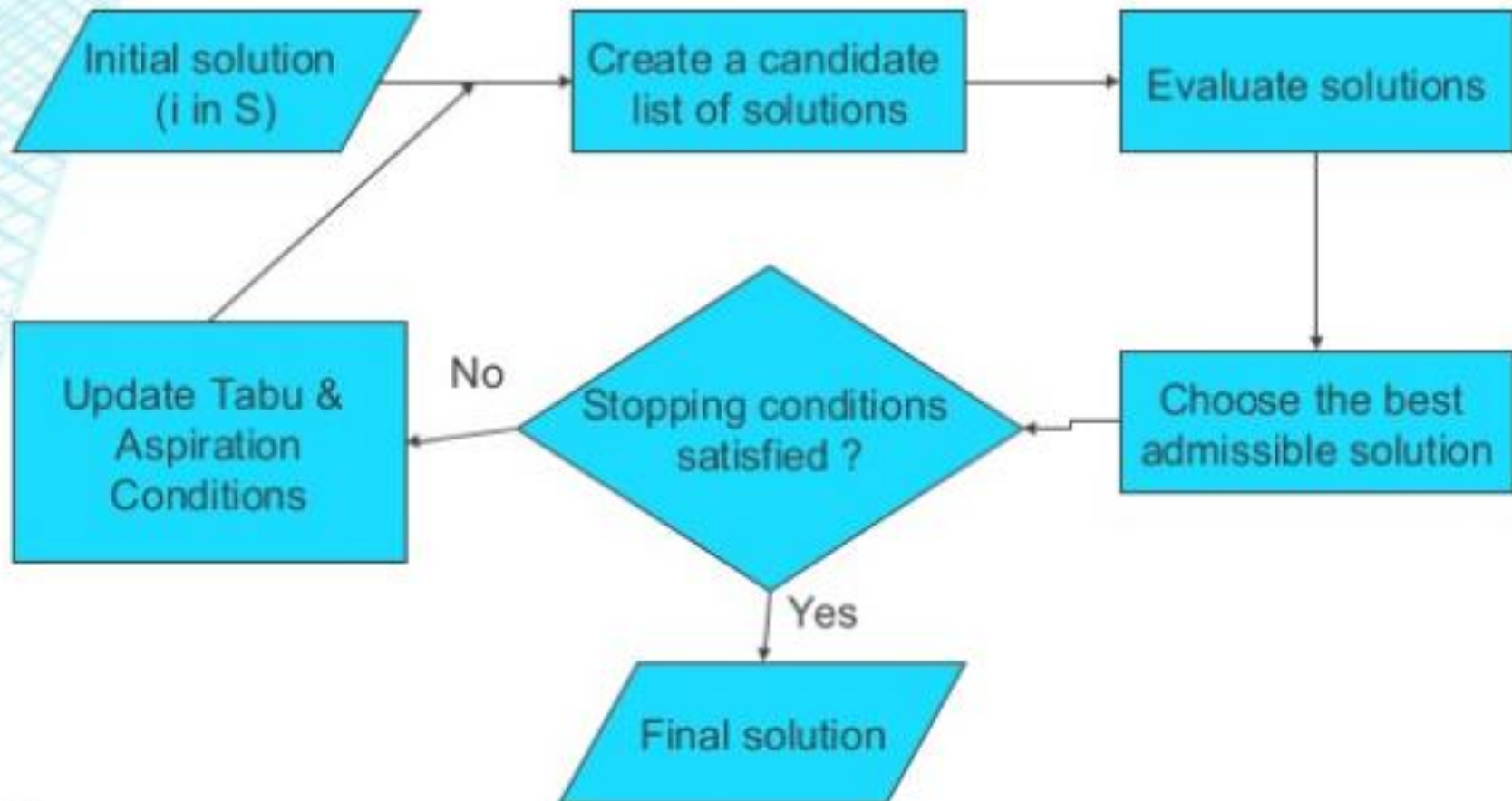
- Step 1: Choose an initial solution i in S . Set $i^* = i$ and $k=0$.
- Step 2: Set $k=k+1$ and generate a subset V^* of solution in $N(i,k)$ such that either one of the Tabu conditions is violated or at least one of the aspiration conditions holds.
- Step 3: Choose a best j in V^* and set $i=j$.
- Step 4: If $f(i) < f(i^*)$ then set $i^* = i$.
- Step 5: Update Tabu and aspiration conditions.
- Step 6: If a stopping condition is met then stop. Else go to Step 2.

Tabu Search Stopping Conditions

Some immediate stopping conditions could be the following :

1. $N(i, K+1) = 0$. (no feasible solution in the neighborhood of solution i)
2. K is larger than the maximum number of iterations allowed.
3. The number of iterations since the last improvement of i^* is larger than a specified number.
4. Evidence can be given that an optimum solution has been obtained.

Flowchart Of Tabu Search Algorithm



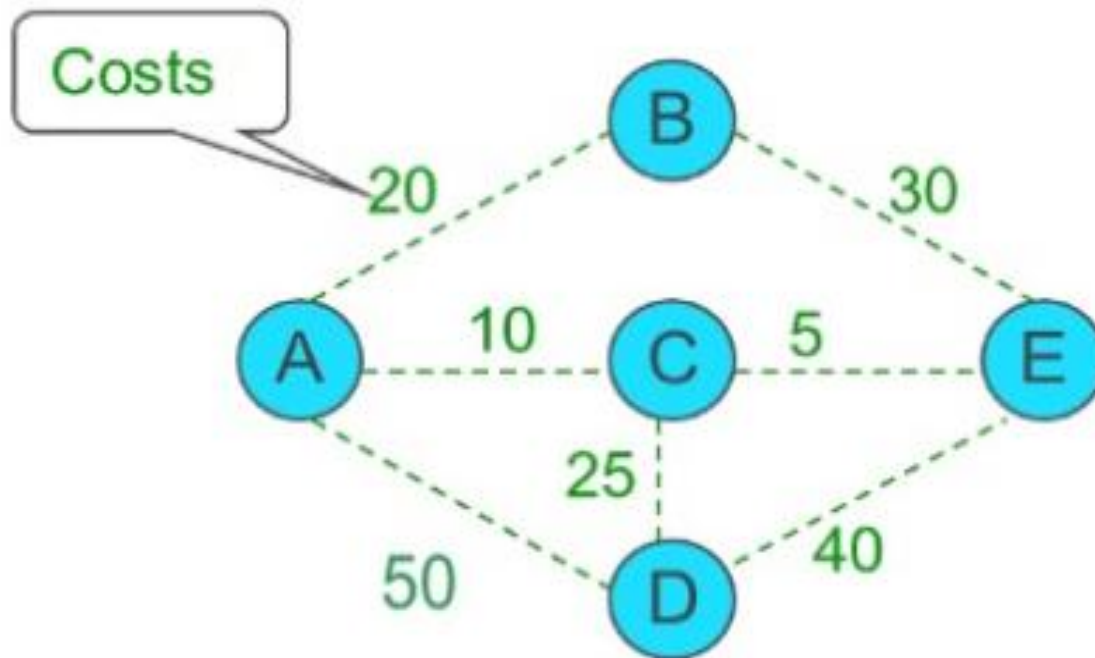
Minimum Spanning Tree

- Given a connected, undirected graph, a spanning tree of that graph is a subgraph which is a tree and connects all the Crests together.
- A single graph can have many different spanning trees. we can also assign a weight to each edge, which is a number representing how unfavorable it is, and use this to assign a weight to a spanning tree by computing the sum of the weights of the edges in that spanning tree. A **minimum spanning tree** or **minimum weight spanning tree** is then a spanning tree with weight less than or equal to the weight of every other spanning tree.

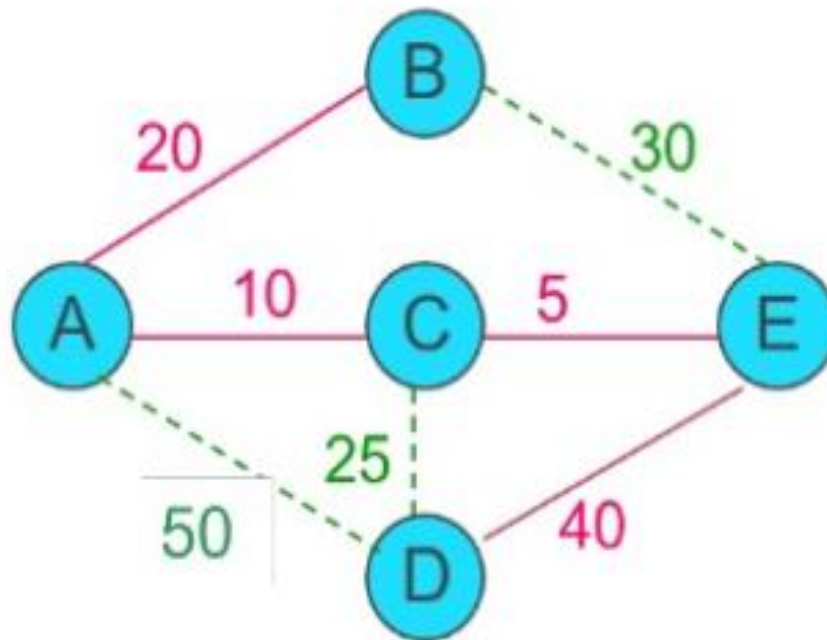
- One example would be a cable TV company laying cable to a new neighborhood. If it is constrained to bury the cable only along certain paths, then there would be a graph representing which points are connected by those paths.
- Some of those paths might be more expensive, because they are longer, or require the cable to be buried deeper; these paths would be represented by edges with larger weights.
- A spanning tree for that graph would be a subset of those paths that has no cycles but still connects. There might be several spanning trees possible. A minimum spanning tree would be one with the lowest total cost.

EXAMPLE

- Minimum spanning tree problem with constraints.
- *Objective: Connects all nodes with minimum costs*



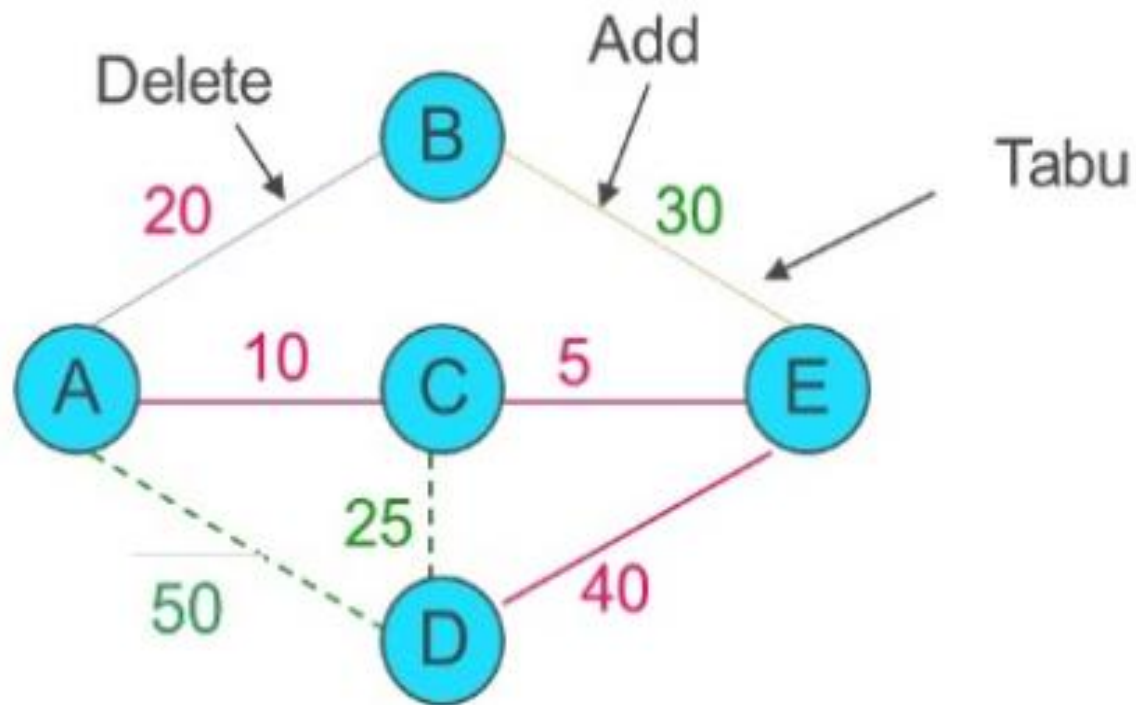
Iteration 1



An optimal solution without considering constraints

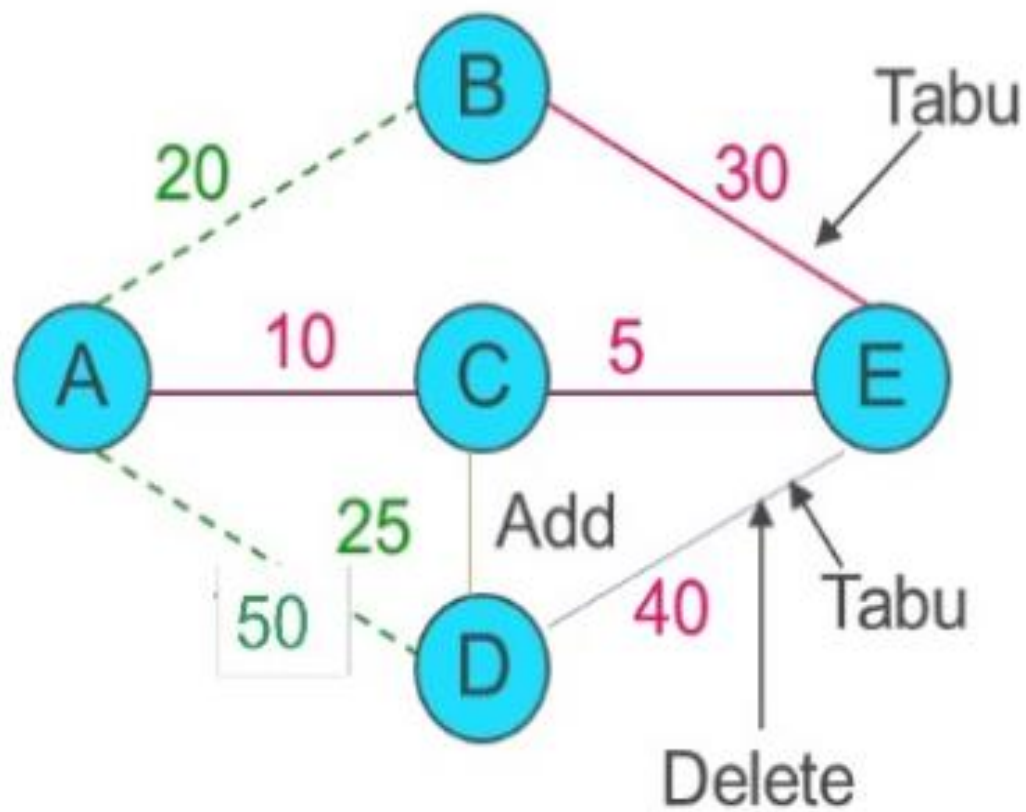
cost = 75 (iteration 1)
(local optimum)

Iteration 2

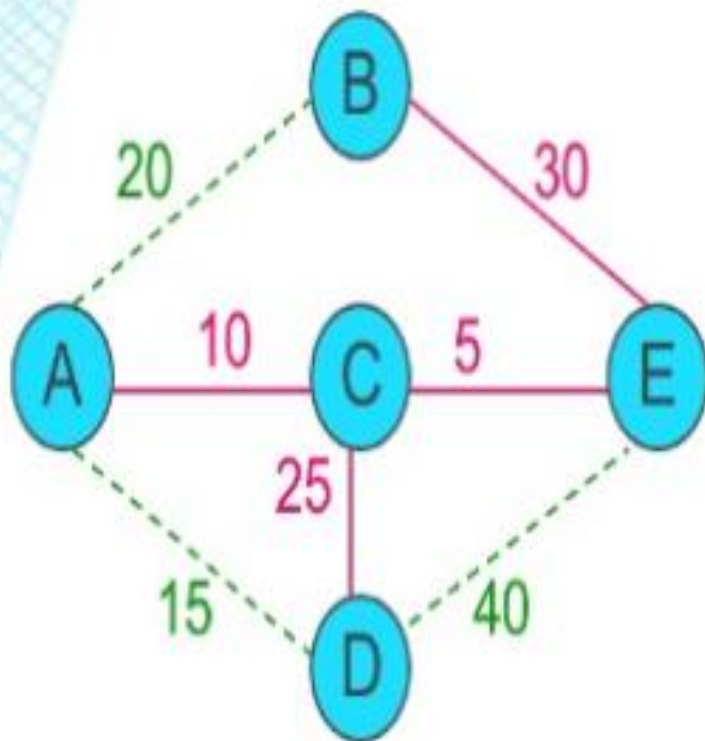


Iteration 2 new cost = 85
Escape local optimum

Iteration 3



Iteration 3 new cost = 70
Override tabu status



Optimal Solution

Cost = 70

Additional iterations only find
inferior solutions

Procedure

- Allows non-improving solution to be accepted in order to escape from a local optimum
- Can be applied to both discrete and continuous solution spaces
- For larger and more difficult problems (scheduling, quadratic assignment and vehicle routing), tabu search obtains solutions that rival and often surpass the best solutions previously found by other approaches.